MTH220 2023 S1 TOA

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* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

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MTH220

Examination - January Semester 2023

Statistical Methods and Inference

Thursday, 25 May 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **EIGHT (8)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Total 100 marks)

Question 1

A health company wishes to release a new "healthy-gut" product range and conducted a market study to understand the demand and interest in their new product. Interviews were conducted for $n = 50$ participants, and an "interest score" between 0 to 100 that represents the level of interest in this product was obtained.

The variables x (age of participant, in years) and y (interest score) are related to each other with the regression equation

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \varepsilon_i \sim N\left(0, \sigma^2\right).$$

Here are some of the reported summary statistics:

$$\sum x^2 = 108066 \qquad \sum y^2 = 314904 \qquad \bar{y} = 76.68$$

Question 1a

Given that the estimated coefficients are $\hat{\beta}_0 = 14.8277$ and $\hat{\beta}_1 = 1.3968$, calculate the values of $\bar{x}$ and

$$\sum xy.$$

Prove that $\bar{x} = 44.28$ and $\sum xy = 183779.5$.

(8 marks)

Question 1b

Calculate the predicted score for $x = 65$. Comment on the $\hat{y}$.

(5 marks)

Question 1c

Suppose, instead of performing a regression of y on x, we perform a regression of y on $(x - 20)$ instead. i.e. The variables x and y are related to each other with the regression equation :

$$y_i = \alpha_0 + \alpha_1 (x_i - 20) + \varepsilon_i, \varepsilon_i \sim N\left(0, \sigma^2\right).$$

Without computing the values of α_0 and α_1 , discuss, compared to the original coefficients β_0 and β_1, will the values of α_0 and α_1 be the same or be different? Explain your reasoning.

(6 marks)

Question 1d

Calculate the Pearson Correlation Coefficient between x and y. Explain how it can be interpreted in the context of the question.

(6 marks)

Question 2a

As part of the regular auditing process, the Quality Control team of a company which produces Jasmine Green Tea canned drinks will measure the value of 100 random samples of the canned drinks every month. The volume, x ml, of a random sample of 100 Jasmine Green Tea canned drinks are summarised as follows:

Sample Mean of x = 328, Sample Variance = $s^2 = 100$. It is known that x follows a normal distribution.

Question 2a(i)

A can of Jasmine Green Tea was stated to be 330ml on the packet. The Quality Control team was tasked to verify if the factory was packaging too little Jasmine Green Tea into the cans. Write down the appropriate hypotheses for this investigation and thus apply a suitable hypothesis test to determine, at the 1% level of significance, whether too little Jasmine Green Tea were packed into the cans. Comment on the results of the hypothesis tests.

(8 marks)

Question 2a(ii)

Using data from the existing sample of 100 canned drinks, compute the probability that a randomly selected canned drink will have a volume less than 330ml.

(3 marks)

Question 2a(iii)

To reduce the number of complaints, the company decides to pack y ml more Jasmine Green Tea per can, such that only 5% of Jasmine Green Tea canned drinks have less than 330ml of green tea. This means that the new cans will have $(x + y)$ ml of Jasmine Green Tea. Using data from the existing sample of 100 canned drinks, find the smallest value of y that meets the condition. Assume that y is a constant value with no variance.

(4 marks)

Question 2b

An insurance company wishes to estimate the average motor insurance claim. Motor insurance claims are well known to be fat-tailed (i.e. it is far more likely to have extreme values compared to normal distribution). It was found that a random sample of 50 claims had a mean claim of $\bar{X} = \$1800$. Assume that the standard deviation of similar claims is $\sigma = \$300$.

Question 2b(i)

Compute the probability for the sample mean of the 50 claims to exceed $1875.

(3 marks)

Question 2b(ii)

Construct a 95% confidence interval for the average claim amount, stating any assumptions you used in your analysis.

(4 marks)

Question 2b(iii)

Determine the required sample size such that the length of the 95% confidence interval is less than $50.

(3 marks)

Question 3

A Physical Proficiency Test was conducted on two groups of students. The test is scored between 0 to 100. It is known that the test results tends to be right skewed, with average around 50. Table Q3 below shows the results of the test.

Group 1					
12	36	93	37	98	64
7	20	69	47	77	57
26	88	58	75	78	56
Group 2					
97	45	13	41	52	58
16	95	77	90	10	54
36	83	38	2	83	16

Table Q3

We want to understand if the 2 groups are different in their physical proficiency.

Question 3a

Suggest a suitable hypothesis test to determine if there is any difference between the two Groups. State the assumptions needed to carry out the test.

(3 marks)

Question 3b

Apply the test at $\alpha = 5\%$.

(13 marks)

Question 3c

Assume now, that you know that the physical proficiency test results come from a right skewed distribution. Discuss the impact of this information on the statistical test you have just performed and state some ways which will help mitigate the issues.

(4 marks)

Question 4a

If we toss a "fair" coin , one for which heads and tails are equally likely, a large number of times, we expect approximately equal numbers of heads and tails. Coins are rarely completely fair- Designs and Engraving could make one side slightly heavier than the other (the heavier side, usually the "head", is more likely to land "face down"); and coins which are bent will land on the convex side more often (more surface area).

Suppose now we want to investigate the fairness of a coin. This will require a lot of coin flips because differences are very minor.

Question 4a(i)

Let X be the number of tails. A coin was flipped 25000 times, and 12520 tails were recorded. Calculate $P(\text{Tails})$ and state the distribution of X and any assumptions made.

(3 marks)

Question 4a(ii)

Show that the distribution of X may be approximated by a normal distribution.

(4 marks)

Question 4a(iii)

Using a suitable approximating distribution, compute the probability we observe between 12450 and 12550 (inclusive) tails in 25000 flips. Remember to apply continuity correction.

(6 marks)

Question 4b

A group of friends want to investigate the probability of getting tails in a coin flip after bending a coin with some pliers. To split the task into more manageable smaller parts, they agreed that they will flip the coin TEN times in each "attempt", and record the number of Tails in the 10 flips for that "attempt". This allows them to do it whenever they have some free time.

After a month, they tallied their numbers. A total of 748 "attempts" were made, each "attempt" made of 10 coin flips for a total of 7480 coin flips. Table Q4(b) below shows the number of Tails in the 748 Attempts.

Let X be the number of Tails obtained in each Attempt.

Number of Tails (i)	Frequency (O_i)
0	4
1	19
2	56
3	164
4	179
5	179
6	68
7	56
8	17
9	3
10	3

Table Q4(b)

Question 4b(i)

Verify that there are a total of 748 Attempts (7480 coin flips) and compute the total number of Tails. Calculate the sample proportion for Tails (total number of tails divided by 7480).

(2 marks)

Question 4b(ii)

Compute and tabulate the expected frequencies assuming X is Binomial. Recall that X is the number of Tails in 10 coin flips.

Number of particle count (i)	Observed Frequency (O_i)	Prob (Assuming Bi) $P(X=i)$	Expected Freq (E_i)
0	4		
1	19		
2	56		
....	...	...	...
10	3		

<u>Table Q4(b)(ii)</u>

(6 marks)

Question 4b(iii)

Apply a suitable test at $\alpha = 5\%$ to verify that X does not follow a binomial distribution (with p = sample proportion).

(7 marks)

Question 4b(iv)

One of the people who did the coin flips said "Since X is a tabulation of findings from a series of coin flips, it should surely be binomial! How can it be otherwise?" Respond to the statement and explain why it is possible for H_0 to be not true, although the process (coin flipping) " should surely be binomial".

(2 marks)

----- END OF PAPER -----

1(a) $B_1 = \frac{S_{xy}}{S_{xx}}$

$B_0 = \bar{y} - B_1\bar{x}$

$n = 50 \quad \Sigma x^2 = 108066 \quad \Sigma y^2 = 314904 \quad \bar{y} = 76.68$

$B_1\bar{x} = \bar{y} - B_0$

$\therefore \bar{x} = \frac{1}{B_1}(\bar{y} - B_0) = 44.28.$

$\Sigma x = n\bar{x} = 2214$

$\Sigma y = n\bar{y} = 3834$

$$B_1 = \frac{S_{xy}}{S_{xx}} = \frac{\Sigma xy - \frac{1}{n}\Sigma x \Sigma y}{\Sigma x^2 - \frac{1}{n}\Sigma x \Sigma x}$$

$\Sigma xy - \frac{1}{n}\Sigma x \Sigma y = B_1(\Sigma x^2 - \frac{1}{n}\Sigma x \Sigma x)$

$\Sigma xy = \frac{1}{n}\Sigma x \Sigma y + B_1(\Sigma x^2 - \frac{1}{n}\Sigma x \Sigma x)$

$= 163779.5$

(b) $\hat{y}(x) = B_0 + B_1 x$

$\hat{y}(65) = 105.1197 \simeq 106$

(c) $y = B_0 + B_1 x$

$y = \alpha_0 + \alpha_1(x - 20)$

$= \alpha_0 + \alpha_1 x - 20\alpha_1$

$= (\alpha_0 - 20\alpha_1) + \alpha_1 x$

Both regression equations should describe the same regression line since they are formulated from the same data

By comparison,

$B_0 = \alpha_0 - 20\alpha_1$

$\alpha_1 = B_1$

(d) $$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{n\Sigma x^2 - \Sigma x \Sigma x}\sqrt{n\Sigma y^2 - \Sigma y \Sigma y}} = 0.967$$

There is strong positive correlation between x and y.

x and y share a linear relationship.

2(c) i. $H_0: \mu \geq 330$
$H_1: \mu < 330$ } Apply lower-tail hypothesis test

population variance unknown, but $s^2 = 100$ $n = 100$

At 1% level significance,
$\alpha = 0.01$
$t_{critical} = qt(\alpha, n-1) = qt(0.01, 99) = -2.36$

$$t = \frac{325 - \mu}{\frac{s}{\sqrt{n}}} = \frac{325 - 330}{\frac{10}{\sqrt{100}}} = -2$$

since $t > t_{critical}$, don't reject H_0.

The amount of Jasmine Green Tea packed are not too little.

ii. $X \sim N(325, 100)$

$P(X < 330) = pnorm(330, 325, \sqrt{100})$
$= 0.5792597$
$\simeq 0.579$

iii. $P(X + y < 330) = 0.05$
$P(X < 330 - y) = 0.05$
$\Rightarrow 330 - y = qnorm(0.05, 325, 10)$
$y = 330 - qnorm(0.05, 325, 10)$
$= 18.44854$
$\simeq 18.4$ ml

2(b) $X \sim N(1800, 300^2)$

$P(X > 1875) = 1 - P(X \leq 1875)$

$= 1 - \text{pnorm}(1875, 1800, 300)$

$= 0.4012937$

$\simeq 0.401$

ii. At 95% confidence level, $\alpha = 0.05$

$\bar{x} = 1800$

$\frac{\sigma}{\sqrt{n}} = 42.42641$

$z_{critical} = \text{qnorm}(1 - \alpha/2) = 1.959964$

upper bound $= \bar{x} + z_{critical} \frac{\sigma}{\sqrt{n}} = 1883.154$

lower bound $= \bar{x} - z_{critical} \frac{\sigma}{\sqrt{n}} = 1716.846$

$\therefore$ confidence interval $= (1883.154, 1716.846)$

iii. length of 95% confidence interval < 50

$2 z_{critical} \frac{\sigma}{\sqrt{n}} < 50$

$z_{critical} = \text{qnorm}(1 - \frac{\alpha}{2}) = 1.959964$

$\sigma = 300$

$\frac{1}{\sqrt{n}} < \frac{50}{2 z_{critical} \sigma}$

$\sqrt{n} > \frac{z_{critical} \sigma}{25}$

$\sqrt{n} > 23.51957$

$n > 553.1701$

$\therefore$ min. sample size $= 554$

3(a) Two-Sample Test for Difference of Two Population Means

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

(b) $n_1 = 18$

$\sum x_{1,i} = 998$

$\bar{x}_1 = \frac{1}{n_1} \sum x_{1,i} = 55.44444$

$s_1^2 = \frac{1}{n_1 - 1} \sum (x_{1,i} - \bar{x}_1)^2 = 767.6732$

$s_1 = 27.70692$

$n_2 = 18$

$\sum x_{2,i} = 906$

$\bar{x}_2 = \frac{1}{n_2} \sum x_{2,i} = 50.33333$

$s_2^2 = \frac{1}{n_2 - 1} \sum (x_{2,i} - \bar{x}_2) = 991.4118$

$s_2 = 31.48669$

$$s_p = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = 879.5425$$

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = 0.0174333 1$$

At $\alpha = 5\%$ with $df = n_1 + n_2 - 2 = 34$

$T_{critical} = qt(1 - \frac{\alpha}{2}, 34) = 2.032245$

since $|T| < T_{critical}$, don't reject H_0

There is insufficient evidence to conclude the 2 groups are different

(c) The statistical test assumes the data is normally distributed, So the test result is no longer correct. To mitigate the situation, we have to transform the data to make it more like a normal distribution. We could apply logarithmic transformation to reduce the skewness of the data.

4(a). i- $P(\text{tails}) = \frac{12520}{25000} = 0.5008$

$X \sim Bin(25000, 0.5008)$

ii. $n = 25000, \ p = 0.5008$

$np = 12700 > 5$
$n(1-p) = 12300 > 5$

$\therefore$ The distribution of X may be approximated by a normal distribution

iii let Y be approximated representation of X

$E(Y) = np = 12700$
$var(Y) = npq = 6248.4$

$\therefore Y \sim N(12700, 6248.4)$

$P(12450 \leq X \leq 12550)$
$= P(X \leq 12550) - P(X \leq 12449)$
$= P(Y \leq 12550.5) - P(Y \leq 12449.5)$
$= \text{pnorm}(12550.5, 12700, \sqrt{6248.4}) - \text{pnorm}(12449.5, 12700, \sqrt{6248.4})$
$= 0.02852841$
$\simeq 0.0285$

4(b) i. total number of attempts $= \sum_{0}^{10} O_i$

$= 4 + 19 + \ldots + 3 + 3$

$= 747$

total number of tails $= \sum_{i=0}^{10} i \cdot O_i$

$= 0 \times 4 + 1 \times 9 + \ldots + 9 \times 3 + 10 \times 3$

$= 3224$

sample proportion $\hat{p} = \frac{3224}{7480} = 0.431016$

ii. Apply $X \sim Bin(10, \frac{3224}{7480})$

$P(X=i) = \binom{10}{i} p^i (1-p)^{10-i}$ where $p = \frac{3224}{7480} = 0.431016$

i	O_i	$P(X=i)$	E_i	$(O_i-E_i)^2/E_i$
0	4	0.00356	2.660	0.67488
1	19	0.02694	20.151	0.06593
2	56	0.09183	68.691	2.34479
3	184	0.18531	138.710	4.23461
4	179	0.24592	183.948	0.013309
5	179	0.22355	167.243	0.83091
6	68	0.14112	105.556	13.36194
7	56	0.06109	45.692	2.32563
8	17	0.01735	12.980	1.24530
9	3	0.00292	2.185	0.30403
10	3	0.00022	0.166	48.54443

4(b) iii. Apply chi-square test for goodness of fit.

$$\chi^2 = \sum_{i=0}^{9} \frac{(O_i - E_i)^2}{E_i} = 74.06235$$

$$df = k - p - 1 = 11 - 1 - 1 = 9$$

$$\chi^2_{critical} = qchisq(0.95, 9) = 16.919$$

Since $\chi^2 > \chi^2_{critical}$, binomial distribution is not a goodfit

iv. Design and engravings could make one side of the coin heavier while coins which are bent will land more often on the convex side. In addition, the coin has been bent prior to the experiment.